



Cost Effectiveness of Allogeneic Umbilical Cord Blood-Derived Mesenchymal Stem Cells in Patients with Knee Osteoarthritis

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Abstract

Objectives The aim of this study was to assess the cost effectiveness of allogeneic umbilical cord blood-derived mesenchymal stem cells with sodium hyaluronate (hUCB-MSC) compared with microfracture in patients with knee cartilage defects caused by osteoarthritis (OA) in South Korea.

Methods A partitioned survival model approach was taken consisting of five mutually exclusive health states: excellent, good, fair, poor, and death over a 20-year time horizon. Utility values were obtained from a randomized clinical trial. Cost data were extracted from a database provided by the Health Insurance Review & Assessment Service, and the utilization of healthcare services was estimated from an expert panel of orthopedic surgeons using a structured questionnaire. The incremental cost-effectiveness ratio (ICER) in terms of quality-adjusted life-years (QALY) was calculated. Deterministic and probabilistic sensitivity analyses were performed.

Results In the base case, the incremental costs of US\$14,410 for hUCB-MSC therapy along with its associated QALY gain of 0.857 resulted in an ICER of US\$16,812 (₩18,790,773) per QALY (95% confidence interval [CI] US\$13,408–US\$20,828) when compared with microfracture treatment from a healthcare payer perspective. From a societal perspective, the ICER was US\$268 (₩299,255) per QALY (95% CI –US\$2915 to US\$3784). When using a willingness-to-pay threshold of US\$22,367/QALY, the probability of hUCB being cost effectiveness compared with microfracture was 99% from the healthcare payer perspective and 100% from the societal perspective.

Conclusions The study demonstrated that hUCB-MSC therapy was cost effective compared with microfracture when treating patients with knee OA. These findings should inform health policy decision makers about considerations for cost-effective therapy for treating knee OA to ultimately enhance population health.

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Key Points for Decision Makers

This study, which is the first cost-effectiveness analysis performed for a stem cell-based drug used to treat knee cartilage defects, found that hUCB-MSC therapy was cost effective for the treatment of knee osteoarthritis when compared with microfracture.

The calculated incremental cost-effectiveness ratio was below the current acceptable maximum willingness-to-pay threshold, which is one times the gross domestic product per capita for South Korea.

The study findings should inform health policy decision makers about considerations for cost-effective therapy for treating knee osteoarthritis to ultimately enhance population health.

1 Introduction

Knee osteoarthritis (OA) is a common condition that affects 240 million people globally and its occurrence is rapidly increasing due to an aging society and rising obesity rates [1]. The global prevalence of symptomatic knee OA was estimated to be 3.8%, with the highest prevalence observed in the high-income Asia Pacific region, which includes countries such as South Korea and Japan [2]. In South Korea, one-third of all adults aged 50 and older are diagnosed with knee OA [3, 4].

Knee OA has a substantial impact on a patient's quality of life (QOL), resulting in worse outcomes compared with that of other chronic diseases or breast cancer [5–9]. Pain and physical disability from knee OA were the two major factors impacting QOL because of the effects on social connectedness, relationships, and emotional well-being [2, 10]. A Korean study also showed knee OA significantly adversely impacted QOL [4].

The nature of knee OA is deteriorative because of its degenerative properties and lack of self-healing potential. The current primary treatment modalities for knee OA include a combination of non-pharmacological and pharmacological interventions [11–13]. When pain cannot be controlled by pharmacological treatments, surgical interventions such as total knee replacement are suggested as an option for end-stage disease management. Multiple approaches to restore the structural and functional properties of cartilage or repair damaged cartilage include use of regenerative medicine and technological advancements including microfracture therapy, autologous chondrocyte

implantation, and mesenchymal stem cell-based therapies, depending on patient profiles [11–15]. Microfracture therapy was developed to treat areas of damaged cartilage (chondral defects) by creating small holes in the bone (i.e., microfractures). This takes advantage of the body's innate healing abilities as the microfractures provide an enriched blood supply and formation of a clot. Subsequently, progenitor cells, which aid in repair after injury, and growth factors form a cartilage layer where the damaged occurred [14–16]. This therapy is often used in patients with low collagen damage and has been proven efficacious for short-term functional improvement but not effective over the long term.

Autologous chondrocyte implantation is commonly used for large lesions and is less often used in the older population due to the diminished reparative potential of the autologous chondrocytes [17, 18]. Autologous chondrocyte implantation is a relatively costly procedure but provides substantial benefits to patients. Many studies found that autologous chondrocyte implantation was a cost-effective therapy compared with microfracture therapy, with some assumptions about long-term outcomes [19–23]. However, a few studies showed favorable cost effectiveness of microfracture therapy compared with autologous chondrocyte implantation [24, 25]. All of these cost-effectiveness studies emphasized the need to explore long-term effectiveness for autologous chondrocyte implantation and microfracture therapy.

A stem cell-based medicinal product, a composite of culture-expanded allogeneic human umbilical cord blood-derived mesenchymal stem cells combined with sodium hyaluronate (hUCB-MSC), has been used for the treatment of knee cartilage defects caused by degeneration from wear and tear and/or repeated trauma in patients with knee OA [18, 26, 27]. hUCB-MSC significantly improved patients' cartilage repair, pain, and physical functions compared with microfracture without causing serious adverse events in short-term and long-term clinical studies [18, 26]. Stem cell-based treatments such as hUCB-MSC could be a promising and effective treatment option for patients with knee OA but a cost-effectiveness analyses of hUCB-MSC has not been conducted.

The objective of this study was to assess the cost effectiveness of hUCB-MSC therapy compared with microfracture in patients who had knee cartilage defects of grade IV, which is based on criteria from the International Cartilage Repair Society (ICRS), caused by degeneration from wear and tear and/or repeated trauma (i.e., OA) from the healthcare payer and societal perspectives in South Korea, respectively, and to provide valuable information to healthcare decision makers when selecting cost-effective therapy for the treatment of knee OA.

2 Methods

2.1 Model Design

A multi-state partitioned survival model over a 20-year time horizon was developed using Microsoft Excel® (Redmond, WA, USA) to assess the potential cost effectiveness of hUCB-MSC therapy compared with microfracture in patients with knee OA from both healthcare payer and societal perspectives in South Korea. Microfracture therapy was chosen as the comparator therapy as it is a commonly used treatment option for knee OA. The model analyzed a hypothetical cohort of patients who were diagnosed with knee OA and were 55 years old at the start of the simulated model [28].

A model schematic illustrating possible transitions between five mutually exclusive health states is shown in Fig. 1. Patients entered the model after the initiation of hUCB-MSC or microfracture and then transitioned from an excellent health state to a good, fair, poor, or death health state. When a patient's health state deteriorated to the poor health state, 18.8% of these patients received total knee replacement surgery based on clinical trial data [18, 26]. Patients were at risk of death from any health state or aging; therefore, they could always move to the death state. One-year cycles were used in the model.

Cost and effectiveness measures were discounted annually at a rate of 5% in accordance with the guidelines recommended by the Korea Health Insurance Review and Assessment Service (HIRA) [28]. A half-cycle correction was applied to improve the accuracy of the results because patients could move along health states at any point during a 1-year cycle.

2.2 Data Source

The effectiveness data used in the model were derived from a clinical trial that evaluated the surgical implantation of hUCB-MSC versus microfracture for treating full-thickness cartilage defects of the femoral condyle in a single compartment in patients with knee pain [18, 26]. Input parameters were based on the intention-to-treat analysis of all participants who were randomly assigned and received one of the surgical procedures described in the study. This clinical trial randomized 114 patients who had a full-thickness femoral cartilage defect of grade IV based on criteria from the International Cartilage Repair Society that was confirmed by arthroscopy or magnetic resonance imaging [18, 29, 30]. The mean age of these patients was 55.3 years for hUCB-MSC and 54.4 years for the microfracture group. The patients who participated in the trial were followed for 5 years. The death rate was estimated using the mortality rate of the general population obtained from age-specific life tables of the 2015 Korean population [31].

2.3 Structure of the Partitioned Survival Model

A partitioned survival analysis extrapolated data derived from the clinical trials and was used to create the cumulative probability survival curves of the five health states over the time horizon of the analysis. The four health states (excellent, good, fair, and poor) were defined using Western Ontario and McMaster Universities Arthritis Index (WOMAC) scores at certain time points during the clinical trial.

The partitioned survival analysis model, which can demarcate the relationships between health states, used parametric survival regressions fitted to Kaplan–Meier curves and generated a survival curve for each health state (excellent, good, fair, poor, and death) using patient-level

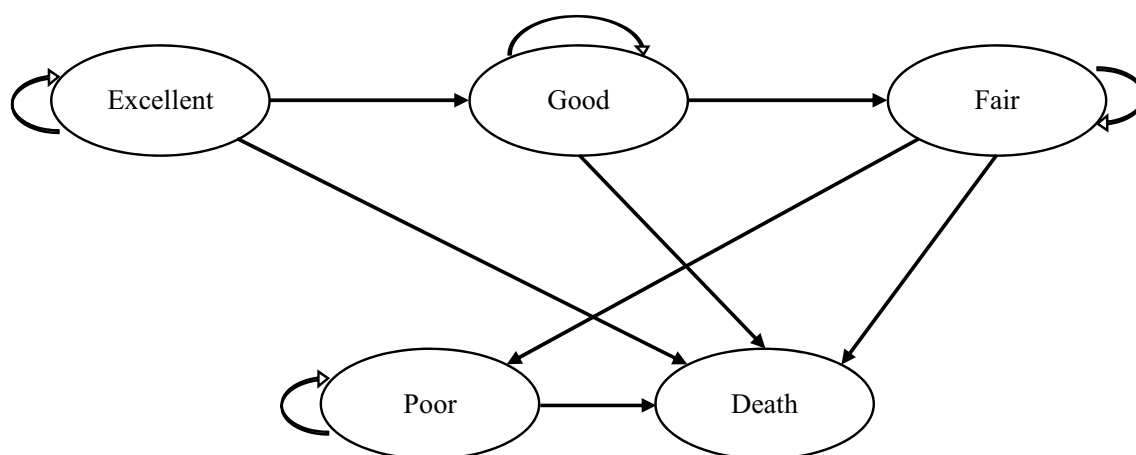


Fig. 1 Partitioned survival model with five health states

clinical trial data. The proportion of patients in each health state at a given time point was estimated from the areas under each health state survival curve [32]. The partitioned survival analysis model has similarities to state transition models but differs in its method of determining the proportion of patients in the health states at each time point [33]. The partitioned survival model was extrapolated over a 20-year time horizon. The first parametric survival curve measured the occurrence of the ‘excellent’ health state over a 5-year period and was extrapolated to estimate the cumulative probability of being in the ‘excellent’ health state according to classification of the WOMAC scores (between 0 and 14 points).

Various distributions (i.e., exponential, Weibull, Gompertz, log-logistic, and log-normal) of the standard parametric models were applied to fit the partitioned survival modeling curves for each health state [34]. The best-fitting independent survival curve was determined using the Akaike Information Criterion and the Bayesian Information Criterion goodness of fit statistics and verified using a visual inspection of the fitted survival curve. The fitted distributions selected for the parametric survival model were a log-logistic distribution for the excellent health state and log-normal for good and fair health states. The selected distributions by health state are shown in Table 1. The fitted survival functions for log-logistic and log-normal are

$$S(t) = 1 + \exp((\log(t) - \mu)/\sigma)^{(-1)}$$

$$S(t) = 1 - \Phi((\log(t) - \mu)/\sigma)$$

where t is time, μ is the scale, and σ is the shape parameter.

The proportion of patients in the ‘excellent’ health state at time t was directly estimated using the survival function $S(t)_1$. The survival function for the ‘good’ health state $S(t)_2$ was estimated sequentially, and the proportion of patients was derived from the difference between $S(t)_1$ and $S(t)_2$. The proportion of patients in the ‘fair’ health state was determined by calculating the difference between $S(t)_2$ and the survival curve for $S(t)_3$. Finally, the proportion of patients in the ‘poor’ health state was calculated as $1 - S(t)_3$. Because knee OA is not a life-threatening condition, the mortality rate of patients with knee OA was assumed to be the same as that of the general population. This model used the mortality rate of the general population obtained from the age-specific life tables of the 2015 Korean population [31]. All statistical and parametric survival analyses were performed using R version 3.5.3 (R Foundation for Statistical Computing, Vienna, Austria) and the flexsurv package [35].

2.4 Utility Values

The degree of joint pain improvement was measured using a visual analog scale, and knee-specific patient-reported outcomes were measured using the WOMAC during clinical trials [36, 37]. The WOMAC is a multidimensional self-administered evaluation instrument that measures clinically important, patient-relevant symptoms in knee or hip OA. This instrument contains 24 questions consisting of three subscales including pain (five items scored 0–4, corresponding to 0–20 points), stiffness (two items scored 0–4, corresponding to 0–8 points), and physical function (17 items scored 0–4, corresponding to 0–68 points), where higher scores reflect worse health status [38]. The number of patients in each health state was calculated using the WOMAC scores collected during a 60-month follow-up period. In the present study, the WOMAC scores were used to classify patient health states into four levels: excellent, good, fair, and poor [18, 26, 39–41]. WOMAC scores from 0 to 14 points corresponded to the ‘excellent’ health state, 15–28 points corresponded to ‘good,’ 29–38 points corresponded to ‘fair,’ and >38 points corresponded to ‘poor’ [39–41].

Utility values for health states were derived from the WOMAC scores collected during the clinical trial [18, 26, 27]. WOMAC scores from the clinical trials were converted to utility index using the EQ-5D mapping algorithm as follows [42]: utility index value = $0.886 - (0.012 \times \text{WOMAC_pain}) + (0.006 \times \text{WOMAC_stiffness}) - (0.009 \times \text{WOMAC_functioning})$. This study used the EQ-5D-3L mapping algorithm to estimate utility values for the health states for consistency because the utility values of total knee replacement were previously measured using EQ-5D-3L [43]. This study applied a mean utility value of 0.717 for patients who received total knee replacement surgery [43]. Quality-adjusted life-years (QALYs) were calculated by multiplying the length of time spent in a particular health state by the utility value of the health state at a certain point in time.

2.5 Resource Use and Costs

This study included costs for hUCB-MSC medication, the initial operation, post-operation, and treatment after recurrence of the poor health state [44]. Medical costs were estimated based on utilized medical services which were collected using a structured questionnaire from an expert panel of seven orthopedic surgeons who performed hUCB-MSC or microfracture surgeries at university hospitals. The questionnaire consisted of questions for hUCB-MSC and microfracture regarding utilization of medical services during initial operation and post-operation (e.g., type, frequency, and

Table 1 Model input parameters for the base-case scenario

Parameters	US\$ (Korea ₩)	Range, US\$	Distribution	Source
Costs associated with operation				
Cost of hUCB-MSC (per vial)	7157 (8,000,000)	(5726–8589)	Gamma	Market price
Cost of medical care with hUCB-MSC therapy	477 (533,693)	(382–573)	Gamma	[44]
Cost of operation for microfracture	363 (406,222)	(291–436)	Gamma	[44]
Cost of medical care with microfracture	435 (486,014)	(348–522)	Gamma	[44]
Cost of post-operation monitoring during hospitalization	404 (451,631)	(323–485)	Gamma	[44]
Cost of monitoring during follow-up (per year)	592 (661,291)	(473–710)	Gamma	[44]
Cost of medication during follow-up (per year)	39 (44,048)	(32–47)	Gamma	[54]
Cost of operation for total knee replacement	5502 (6,150,000)	(4402–6603)	Gamma	[44]
Cost of monitoring after total knee replacement	823 (919,881)	(658–988)	Gamma	[44]
Time costs				
Cost of outpatient visit for operation	21 (23,762)	(17–26)	Gamma	[44]
Cost of outpatient visit after operation	32 (35,643)	(26–38)	Gamma	[44]
Cost of transportation per visit	9 (10,228)	(7–11)	Gamma	[47]
Cost of caregiver per day	89 (100,000)	(72–107)	Gamma	Market price
Productivity loss				
Patient wage per hour	14 (15,133)	(11–16)	Gamma	[46]
Days of absenteeism in excellent state (per year)	1.5	(1.2–1.8)	Normal	Panel survey
Days of absenteeism in good state (per year)	6.2	(5.0–7.4)	Normal	Panel survey
Days of absenteeism in fair state (per year)	15.7	(12.5–18.8)	Normal	Panel survey
Days of absenteeism in poor state (per year)	36.7	(29.3–44.0)	Normal	Panel survey
Proportion of presenteeism in excellent state	0.05	(0.04–0.06)	Gamma	Panel survey
Proportion of presenteeism in good state	0.20	(0.16–0.24)	Gamma	Panel survey
Proportion of presenteeism in fair state	0.50	(0.40–0.60)	Gamma	Panel survey
Proportion of presenteeism in poor state	0.88	(0.70–1.00)	Gamma	Panel survey
Utility values for health state (mean, 95% CI)				
Excellent state	0.825	(0.815–0.835)	Beta	[18, 27]
Good state	0.716	(0.704–0.727)	Beta	[18, 27]
Fair state	0.609	(0.599–0.620)	Beta	[18, 27]
Poor state	0.413	(0.387–0.440)	Beta	[18, 27]
Survival function*				
	Scale (μ)	Shape (σ)		
hUCB-MSC therapy				
Excellent state $S(t)_1$	15.239	2.639	Log-logistic	[18, 27]
Good state $S(t)_2$	4.237	1.326	Log-normal	[18, 27]
Fair state $S(t)_3$	5.049	1.467	Log-normal	[18, 27]
Poor state $S(t)_4$	$1 - S(t)_3$			[18, 27]
Microfracture				
Excellent state $S(t)_1$	14.907	2.883	Log-logistic	[18, 27]
Good state $S(t)_2$	3.413	1.038	Log-normal	[18, 27]
Fair state $S(t)_3$	3.918	1.114	Log-normal	[18, 27]
Poor state $S(t)_4$	$1 - S(t)_3$			[18, 27]

Korean won (₩) were converted to United States dollars (US\$) using the average currency exchange rate of ₩1 = US\$0.00089467 between 1 January and 30 June 2021

CI confidence interval, hUCB-MSC human umbilical cord blood-derived mesenchymal stem cells

*The survival function for the ‘excellent’ health state was estimated using a log-logistic distribution, and survival functions for the ‘good’ and ‘fair’ health states were estimated using log-normal distribution

duration of medical services for outpatient, hospitalization, and physical therapy, respectively) and medication use (e.g., dosage, strength, and duration of medications, incidence and treatment of adverse effects). This study assumed that two vials of hUCB-MSC (2.5 mL) were administered based on the results of the phase III clinical trials, and the market price per vial was US\$6735 (₩8,000,000 Korean won).

Initial operation costs included those associated with hUCB-MSC, microfracture surgery, hospital stay, outpatient visits, and post-operation treatment and monitoring (including physical therapy) during hospitalization. The average hospital stay was 9 days for patients treated with hUCB-MSC and 8 days for patients treated with microfracture based on surveying the expert panel of seven orthopedic surgeons using structured questionnaires. Post-operation costs included the annual costs of outpatient visits, which were assumed to be once a month, monitoring, physical therapy follow-up, and post-operation medication. After patients received total knee replacement surgery, the treatment costs included the costs of operation, monitoring, medication, and physical therapy, which was assumed to be 20 times per month for 3 months following the operation.

Those unit costs were retrieved from the Korean Medical Service Fee, and the price of drugs was listed by the HIRA [48]. The treatment costs of total knee replacement surgery were retrieved from the nationwide average cost reported by the HIRA. All costs were adjusted to 2021 Korean won (₩) based on the medical cost component of the consumer price index. These costs were then converted to 2021 United States dollars (US\$) using the average currency exchange rate between January 1 and June 30, 2021 (₩1 = US\$0.00089467). Cost inputs in the model are presented in Table 1.

2.6 Time and Productivity Costs

When the cost-effectiveness analysis was performed from the societal perspective, the model included time and productivity costs which included costs associated with treatment, such as time spent by patients, their families, and caregivers; costs associated with lost or impaired ability to work or to enjoy leisure activities due to morbidity; and economic productivity loss due to death [45]. Costs associated with patients and caregivers were based on length of time spent on treatment-related activities/services collected from an expert panel of seven orthopedic surgeons, and wage data were derived from reports of government statistics (Table 1).

Time costs related to receiving treatment were estimated based on the mean hourly wage of labor conditions by employment type [46], and the mean time spent for outpatient visits and hospitalization was based on a survey examining national medical care resources and utilization. The average labor wage was calculated by applying a weighted

average to the age distribution of trial participants multiplied by the time spent for treatment. Caregiver time costs were calculated by multiplying the number of hospitalization days by the market labor wages per day for caregivers. Transportation costs for clinic visits (i.e., direct nonmedical costs) were calculated using the Third Korea National Health and Nutrition Examination Survey [47].

Absenteeism, which measures the number of days patients were absent from work per year, and presenteeism, which measures the proportion of reduced work performance, affected by knee OA were indirectly measured by a clinical expert using the Work Productivity and Activity Impairment Structured questionnaire (general health V2.0) [48]. Costs associated with work productivity loss stratified by health state and type of knee operation were calculated using the following equation: costs associated with work productivity loss = ([number of days absent from work/total number of yearly working days] + [number of working days/total number of yearly working days × proportion of impaired ability to work]) × (average labor wage).

2.7 Sensitivity Analyses

The robustness of the study results was tested using one-way sensitivity and probabilistic sensitivity analyses.

2.7.1 One-Way Deterministic Sensitivity Analysis

A one-way sensitivity analysis was performed to measure the relative impact of key parameters on the cost effectiveness of hUCB-MSC treatment. Parameters tested for model outcomes included discount rates (ranging between 0% and 7.5% for cost and effectiveness) and utility values (varied using 95% confidence intervals [CIs] of change in utilities for each health state). The model also varied costs using a range of $\pm 20\%$ of the point estimates, which included medical test costs per year after surgery, the cost of total knee replacement, and the cost of hUCB-MSC.

2.7.2 Probabilistic Sensitivity Analysis

Probabilistic sensitivity analyses were performed with 10,000 Monte Carlo simulations to assess the joint impact of key input parameter uncertainties on the cost effectiveness of hUCB-MSC therapy. The assumptions and distributions for the input parameters are listed in Table 1. Utilities were varied using the beta distribution, and cost parameters were varied using gamma distributions [49].

The model calculated the cumulative costs and QALYs for 20 years following the year of the surgery using a 5% discount rate. The ICERs and their respective 95% CIs between hUCB-MSC and microfracture were calculated. Cost-effectiveness acceptability curves were constructed

over a wide range of willingness-to-pay thresholds to determine the probability that hUCB-MSC treatment was a cost-effective therapy.

3 Results

The base-case analysis comparing hUCB-MSC therapy with microfracture is presented in Table 2. An additional 0.857 QALYs were gained with the hUCB-MSC therapy (8.256 QALYs) compared with microfracture (7.399 QALYs). From a healthcare payer perspective, the total cost per patient was US\$23,831 for the hUCB-MSC therapy and US\$9421 for microfracture. These cost and health outcomes resulted in an ICER of US\$16,812 (₩18,790,773) per QALY gained (95% CI US\$13,408–US\$20,828). From a societal perspective, the total cost per patient was US\$65,623 for the hUCB-MSC therapy and US\$65,394 for microfracture. The incremental costs of hUCB-MSC therapy compared with microfracture were US\$229 (₩256,498), with an associated gain of 0.857 QALYs, resulting in an ICER of US\$268 (₩299,255) per QALY (95% CI –US\$2915 to US\$3784) in the base-case analysis.

The one-way sensitivity analysis presents the eight variables that had the most influential effects on the ICER values out of the key study variables (Fig. 2). From the healthcare payer perspective, the duration of analysis was the most influential parameter that caused ICER value change, followed by the discount rate for effectiveness. However, from the societal perspective, the discount rate for cost was the most influential model parameter, followed by reduced productivity of presenteeism in the poor health state.

The uncertainty around ICERs was plotted on the cost-effectiveness plane using Monte-Carlo simulations (Fig. 3). As shown in the cost-effectiveness planes from the societal and healthcare payer perspectives, all ICER simulations were located in the right-hand quadrants for both perspectives, suggesting hUCB-MSC therapy was potentially cost effective or dominant. When using a willingness-to-pay threshold of US\$22,367/QALY, the probability of hUCB being cost effective compared with microfracture was 99% from the healthcare payer perspective and 100% from the societal perspective (Fig. 4).

Table 2 Cost-effectiveness results for the base-case analysis

Parameters	hUCB-MSC (US\$)	Microfracture (US\$)	Incremental analysis	
			US\$	Korean won (₩)
Costs				
Cost of initial operation*	15,156	798	14,357	16,047,679
Cost of post-operation treatment*	8113	7993	120	134,282
Cost of treatment after recurrence*	562	630	−68	−75,970
Cost of transportation	369	364	5	5802
Cost of caregiver time	557	545	11	12,716
Cost of patient time	2309	2162	147	163,981
Cost of work productivity loss	38,558	52,901	−14,343	−16,031,992
Total costs from healthcare payer perspective*	23,831	9421	14,410	16,105,991
(95% CI)	(20,797–27,199)	(8047–10,893)	(11,680–17,493)	(13,055,585–19,552,519)
Total costs from societal perspective	65,623	65,394	229	256,498
(95% CI)	(62,589–68,991)	(64,020–66,866)	(−2500 to 3313)	(−2,793,908 to 3,703,026)
QALYs	8.256	7.399	0.857	0.857
(95% CI)	(8.157–8.355)	(7.241–7.562)	(0.774–0.941)	(0.774–0.941)
ICER from healthcare payer perspective			16,812/QALY	18,790,773/QALY
(95% CI)			(13,408–20,828)	(14,986,350–23,279,609)
ICER from societal perspective			268/QALY	299,255/QALY
(95% CI)			(−2915 to 3784)	(−3,257,870 to 4,231,519)

Korean won (₩) were converted to United States dollars (US\$) using the average currency exchange rate between 1 January to 30 June 2021 (₩1 = US\$0.00089467)

CI confidence interval, hUCB-MSC human umbilical cord blood-derived mesenchymal stem cells, ICER incremental cost-effectiveness ratio, QALY quality-adjusted life-year

*These costs were included to calculate the total costs from the healthcare payer perspective

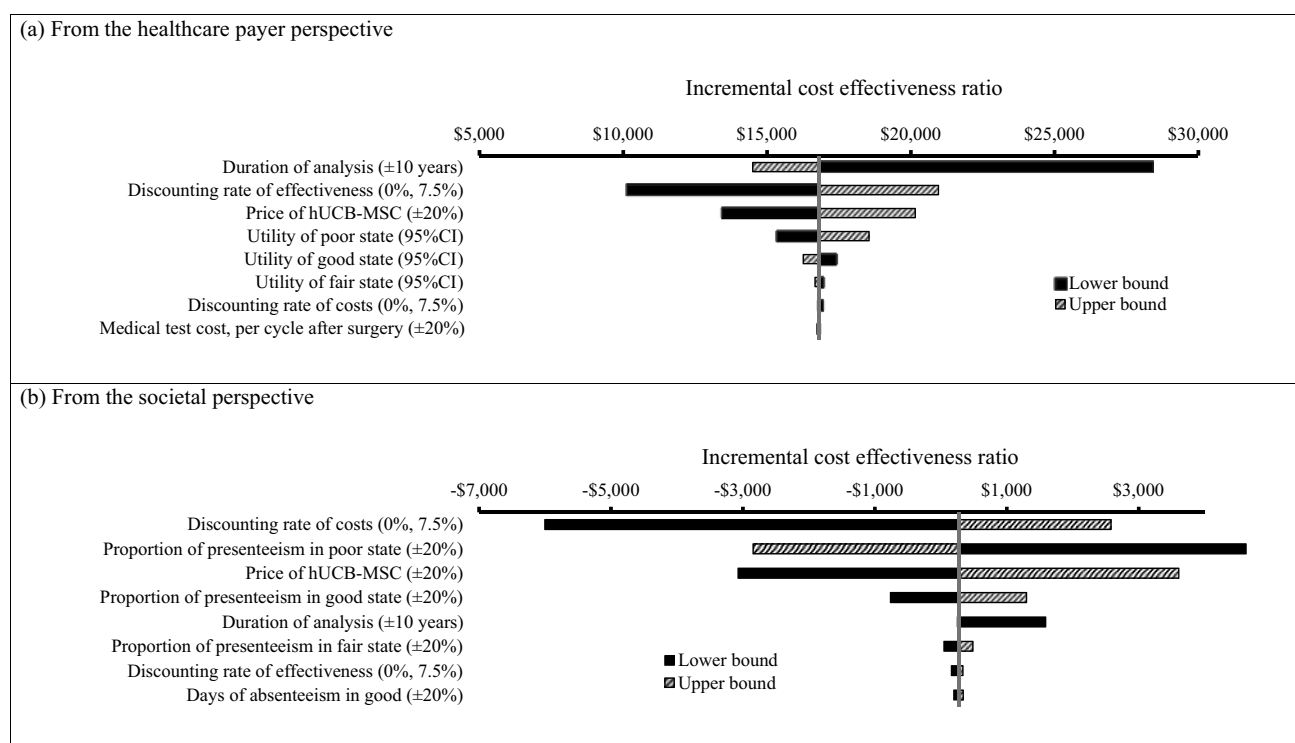


Fig. 2 Tornado diagram of a one-way sensitivity analyses for hUCB-MSC therapy compared with microfracture. *hUCB-MSC* human umbilical cord blood-derived mesenchymal stem cells

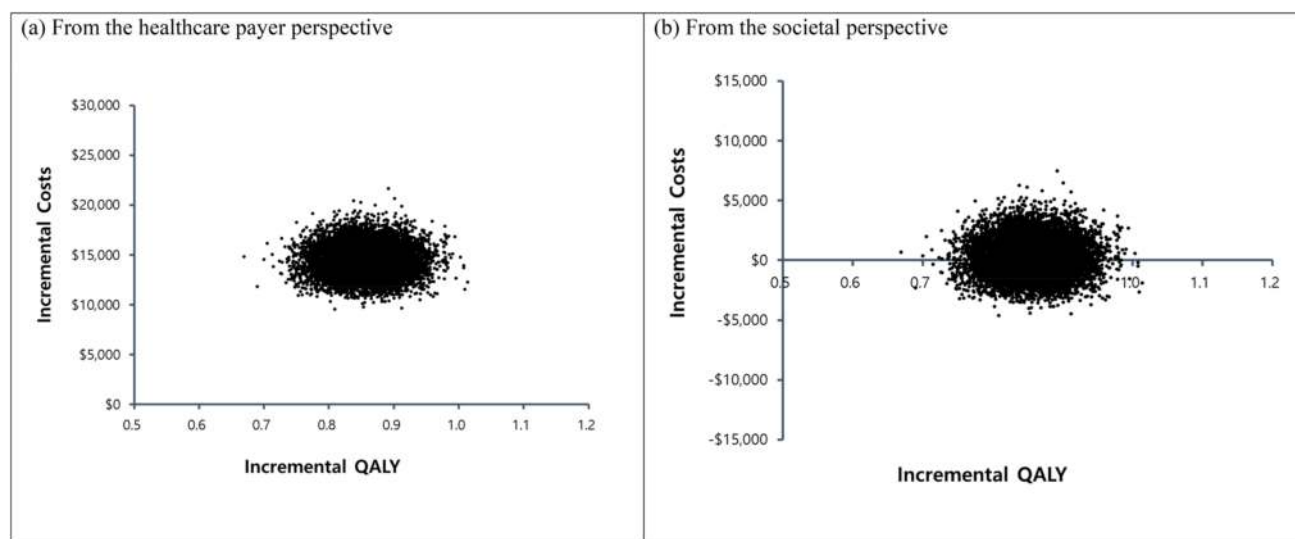


Fig. 3 Cost effectiveness scatterplot of hUCB-MSC therapy compared with microfracture. *hUCB-MSC* human umbilical cord blood-derived mesenchymal stem cells

4 Discussion

To the best of our knowledge, this is the first cost-effectiveness analysis performed for a stem cell-based drug

used to treat knee cartilage defects using 5-year long-term follow-up data from a randomized clinical trial. This study found that hUCB-MSC therapy was cost effective, with an ICER per QALY gained of US\$268 (₩299,255) from the societal perspective and US\$16,812 (₩18,790,773)

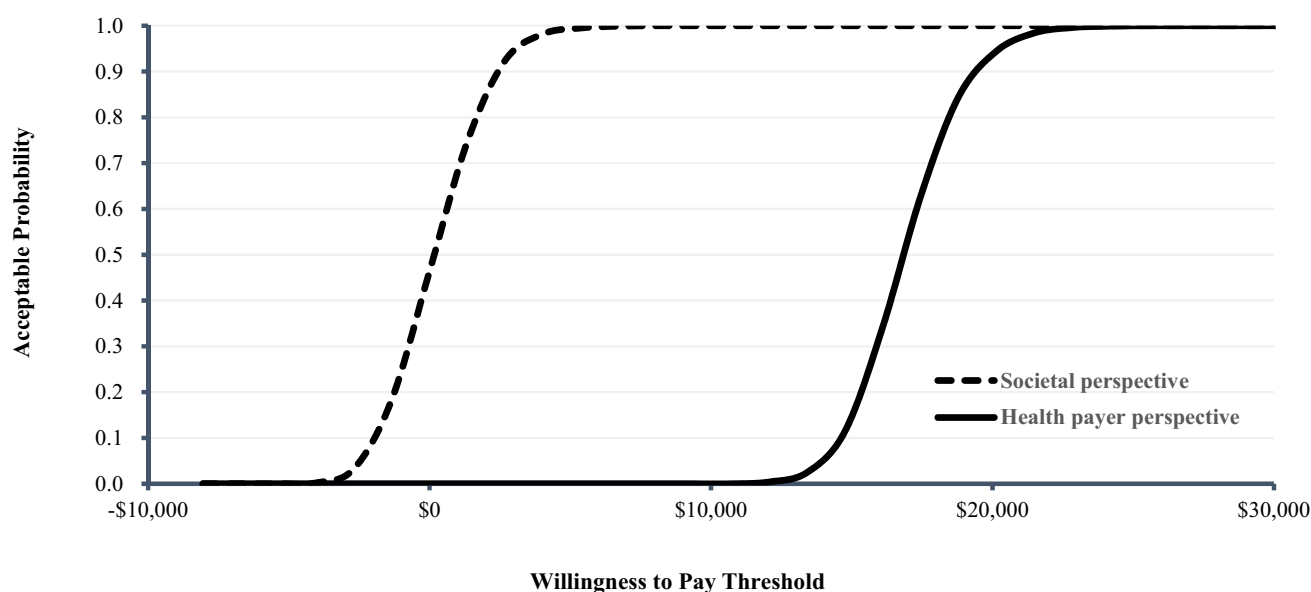


Fig. 4 Cost-effectiveness acceptability curves of hUCB-MSC therapy versus microfracture. *hUCB-MSC* human umbilical cord blood-derived mesenchymal stem cells

from the healthcare payer perspective for the treatment of knee OA when compared with microfracture. This ICER was below the acceptable maximum willingness-to-pay threshold for a new drug (except for cancer) of US\$22,367 (₩25,000,000) per QALY, as commonly used by the HIRA in Korea. The differences in ICER between the two perspectives were attributable to a much lower productivity loss with the hUCB-MSC therapy. Although HIRA requested a cost-effectiveness analysis from the healthcare payer perspective, this study conducted analyses from both the healthcare payer and societal perspectives to help readers compare the results of this study to those of other studies [50].

Most previous studies assessed or summarized cost effectiveness of autologous chondrocyte implantation compared with a common comparator such as microfracture, which was the prevailing therapy for the treatment of cartilage defects of the knee prior to the introduction of hUCB-MSC therapy [15, 19–25]. The ICER values of autologous or characterized chondrocyte implantation compared with microfracture (using USD currency and the US medical service price index in 2021) ranged from US\$25,251/QALY to US\$81,600/QALY depending on scenarios [20–22]. Cost-effectiveness analyses using a decision tree model with a 40-year time horizon from the global healthcare payer perspective in Belgium found that the ICER value was US\$25,251/QALY (as of 2021) [20]. A study conducted using a decision tree model with a duration of 47 years from the German statutory health insurance perspective estimated ICER values ranged from US\$58,656 to US\$81,600 per QALY (as of 2021) [21]. A cost-effectiveness study

of characterized chondrocyte implantation compared with microfracture estimated the ICER values to be US\$32,929/QALY (as of 2021) when a Markov model was applied for a 75-year time horizon from the UK National Health Service perspective [22]. In contrast, a review study reported that microfracture was the most effective treatment option compared with autologous chondrocyte implantation for the chondral lesions in the knee; this finding was later supported by another review study [24, 25]. Both review studies recommended well designed clinical trials with an appropriately long-term time duration to ensure robust outcomes.

In general, the range of ICER values found for autologous chondrocyte implantation from the literature was slightly higher than those found in the present study. This difference can be partially explained by the clinical characteristics of autologous chondrocyte implantation, which requires two arthroscopic procedures: one to collect the cells and one to implant them. If the most recent autologous chondrocyte implantation method, a one-step procedure, was applied, the ICER values would likely be lower.

The present study had several strengths, including the use of long-term outcome data collected during 5-year follow-up clinical trials, and a partitioned survival model that constructed survival curves from Kaplan–Meier curves, which the present study used to construct health states [18, 33]. Therefore, long-term efficacy, utility, and safety data were available to establish a comprehensive partitioned survival model. The survival functions that inform patient proportion estimates in the partitioned survival analysis make the models intuitively appealing, easy to construct and use to communicate results, and able to be replicated using in-trial

data [33]. The time frame of this study was 20 years (i.e., life-time horizon) with patients who were 55 years old at the start of the model. Because knee OA is a disease that persists throughout an individual's life, the lifetime horizon used for the model was appropriate, and a similar time frame was also observed in previous cost-effectiveness studies of autologous chondrocyte implantation for cartilage defects [20, 22, 23, 51].

The present study estimated the cumulative probabilities between the four health states with a partitioned survival analysis using actual WOMAC scores that were measured annually to assess the efficacy and functional status during 5-year clinical trials [18, 26]. Using these four health states and the death state throughout the model analyses was one of the study's strengths because these states reflected detailed post-operation outcomes during the 5-year long-term follow-up period. The cut-off points of WOMAC scores for each of the health states corresponded to the pre-existing classification of severe arthritis, defined as a WOMAC score of 39 or greater [39–41]. A recent study suggested that the four health states of WOMAC scores can also be classified based on the level of patient satisfaction after total knee arthroplasty [52].

In the present study, the costs considered from a healthcare payer perspective were similar to previous cost-effectiveness studies of autologous chondrocyte implantation that used a Markov model [22, 23]. The cost of microfracture was estimated to be between US\$4092 and US\$5523, including the costs of the procedure, inpatient stay, outpatient visits, and rehabilitation [22, 23]. The present study assumed that hUCB-MSC required 9 inpatient days and microfracture required 8 days, and outpatient visits for both therapies required two visits before the surgery and three visits annually after the surgery. These assumptions were based on data collected from an expert panel of orthopedic surgeons. Despite the fact that the frequency of inpatient or outpatient visits for the procedure could be slightly varied, the cost of microfracture therapy was assumed to be US\$798 (₩892,236).

Although this study provides valuable information on the cost effectiveness of hUCB-MSC therapy for patients with knee OA using long-term follow-up data collected in a randomized clinical trial, the findings should be interpreted carefully because of limitations. First, because healthcare utilization, productivity loss, and effectiveness data were derived from a clinical trial conducted in Korea, treatment costs may need to be adjusted for each country's healthcare delivery and health insurance systems. Thus, the generalizability of these results to other countries should be interpreted with caution.

The use of clinical trial data might cause some inherent bias for the effectiveness of treatments. The costs may not have been estimated accurately because of factors such as

the evaluation of patients' healthcare utilization patterns and proxy of work productivity losses using structured questionnaires filled out by clinical experts. Utility values were converted from WOMAC scores using the mapping algorithm of EQ-5D, which was developed based on the UK population, because a mapping algorithm for the Korean population was not available. Utility values were also assumed to be the same for both the healthcare payer and societal perspectives because of limited data. However, performing deterministic and probabilistic sensitivity analyses may have addressed some of these uncertainties. Also, the main limitation of the partitioned survival model is its fundamental structural assumption that the survival endpoints are independent. Assuming independence between endpoints reduces the value of a partitioned survival model as a means of quantifying decision uncertainty [33]. A previous study that used a partitioned survival analysis and Markov models produced very similar expected costs, outcomes, and ICERs [53].

5 Conclusion

This study found that hUCB-MSC therapy for patients with knee OA was cost effective when compared with microfracture from both the healthcare payer (ICER = US\$18,790/QALY gained) and societal perspectives (ICER = US\$268/QALY gained). Patients and healthcare providers may consider hUCB-MSC therapy an economically attractive treatment option compared with microfracture, addressing an unmet need for improving overall outcomes. This study provides a framework for healthcare payers and practitioners to assess the cost effectiveness of stem cell-based medical products for treating cartilage regeneration in patients with knee OA. The findings should inform health policy decision makers about considerations for cost-effective therapy for treating patients with knee OA to enhance patient QOL. Future research should explore the long-term cost effectiveness of hUCB-MSC therapy with less uncertainty regarding the study parameters.

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Declarations

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Conflicts of interest The authors declare that they have no conflict of interest to disclose.

Availability of data and materials The datasets used and/or analyzed during the present study are available from the corresponding author upon reasonable request.

Code availability The model used for the present study is available from the corresponding author upon reasonable request.

Author contributions KS contributed to the study conception and design, data collection, analysis and interpretation of the data, and manuscript development. BJC, AG, SML CWH, DCS, and MKK participated in data collection and analysis and writing the draft manuscript. DCS, HC, GYH, and SML performed the cost-effectiveness analysis and simulation. KS and SML had full access to all the data in the study and take responsibility for the integrity of the work as a whole. All authors contributed to the final manuscript's writing, editing, and approval.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication (from patients/participants) Not applicable.

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